

AMD AI DevMaster Hackathon 2026 • Track 1

MiniMax H3 • Local Multimodal Video Studio

Text / images → video + **native stereo audio** on a single AMD Radeon GPU

ROCm 7.2.4

Radeon PRO W7900

pruned int8

100% local

Why this tool

Pain point

- Video + audio generation needs cloud APIs or heavy NVIDIA stacks
- Open-weights models (bf16) are 190 GB+ — won't fit consumer GPUs
- Most tools do video or audio, rarely both, natively

Our answer

- Omni-modal MiniMax H3: one sampling pass renders **video + audio**
- **pruned int8** set fits 48 GB VRAM on Radeon
- 100% local & private — no cloud calls

Creators • agencies • e-learning • IP studios • privacy-sensitive teams

Three generation modes

T2V

text prompt → video + stereo audio

Cinematic shots, narration, ambience

I2V

prompt + first/last keyframes → video + audio

Shot-constrained animation

R2V

prompt + reference images → identity-locked video + audio

Character / style reuse via <Picture 1>

Audio is native, not layered on: voice, sound effects and music are modeled jointly with the video in a single forward pass.

System architecture

```
gradio/streamlit UI → ComfyUI API (8188) → MiniMaxH3 nodes  
  → SamplerCustomAdvanced (joint video+audio latents)  
  → VAEDecode (video) + VAEDecodeAudio (audio) → CreateVideo → mp4
```

Client — `comfy_client.py` builds prompt graphs, uploads keyframes, polls history, returns the mp4.

Server — ComfyUI native nodes on ROCm PyTorch; no CUDA code paths.

Models: pruned int8 fl2va/ref2va (~21 GB) • int8 Qwen3-VL-32B (27 GB) • fp16 video VAE + fp32 audio VAE

Generated outputs



Every clip ships with **embedded stereo audio** generated by the model. 9 clips in `assets/showcase/` ; full demo video in `assets/demo.mp4` (2:34 @ 1080p).

AMD Radeon / ROCm adaptation

Choice	Why
pruned int8 diffusion (fl2va/ref2va)	bf16 is 66 GB / file — pruned set removes training-only weights, int8 shrinks to ~21 GB
int8 Qwen3-VL-32B text encoder	27 GB, native on RDNA (no FP4 emulation)
fp16 video VAE + fp32 audio VAE	small, full precision where it matters
ComfyUI native nodes	same nodes run on ROCm PyTorch (HIP) — no CUDA translation
20 steps • res_multistep • simple	Comfy-Org reference config, quality/speed balance

Measured performance

Mode	Resolution	Duration	Wall time
T2V (cold)	864x480	~5.2 s	10:42 (incl. model load)
T2V (warm)	608x352	~2.3 s	75 s
T2V (warm)	864x480	~2.3 s	170 s
I2V (warm)	864x480	~5.2 s	553–617 s

Honest disclosure: image-conditioned modes (I2V/R2V) take ~9–10 min for a 5 s clip — conditioning tokens ride every sampling step. We report this openly rather than only showing best-case T2V numbers.

Reproducibility

```
git clone https://github.com/Comfy-Org/MiniMax-H3 ✓ hf download (pruned int8)
python main.py --listen 0.0.0.0 --port 8188    # ComfyUI
streamlit run app/streamlit_app.py             # web studio
python app/bench.py --all --skip-queue         # validate vs live server
python app/test_comfy_client.py               # 8 unit tests, no GPU
```

Unit tests (8) — graph construction, 17k+5 frame grid, mode wiring

Benchmark script — all modes, wall time + peak RSS, JSON output

Use cases & practical value

- **Short-form social video** — clips with synced sound in one shot
- **Ads & storyboards** — pitch concepts with audio ambience
- **E-learning explainers** — localized narration in-loop
- **Game cinematics / IP** — style-locked R2V character reuse
- **Privacy-safe creation** — everything stays on-device

Differentiator

Among Track 1 video tools, this is the one that generates **video and native synchronized audio together** — voice, SFX and music come out of the same diffusion pass, so lip-sync and ambience are naturally aligned.

Thank you

MiniMax H3 • Local Multimodal Video Studio on AMD Radeon

Team peace_&_love • Track 1 • AMD AI DevMaster Hackathon 2026

PR #90

video + native stereo audio

ROCm • RDNA3